

Chaitanya Madisetty Data Engineer

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Summary

A **Senior Data Engineer** with **5+** years of experience in **Cloud Infrastructure** and **Advanced Big Data Analytics** within large-scale Data and various IT Environments. Expert in utilizing **Apache Nifi, Talend, Azure Data Factory, Databricks, Synapse Analytics, Snowflake Matillion, Snowflake DBT, AWS Glue, and AWS EMR** to architect and manage scalable **ETL pipelines**, leading to substantial improvements in data processing efficiency. Skilled in migrating **on-premises databases** from **Oracle, Microsoft SQL Server, MySQL, PostgreSQL, Cassandra, and Teradata** to **cloud platforms** such as **Azure, Snowflake, and AWS**. Executing complex/large data transformations to deliver actionable business insights. Proficient in **data mining, modeling, data warehousing, and real-time data streaming** using **Erwin, Apache Kafka, Spark, Azure Stream Analytics, and Azure Event Hubs**, and experienced in leveraging **REST and SOAP APIs** for data integrations and pipeline scaling. Adept at developing interactive dashboards with **PowerBI and Tableau**. Experienced in applying **Machine Learning Algorithms** such as **Regression, Classification, and Clustering** for **predictive analytics** and deploying models in a production environment using **TensorFlow, MLlib, MLflow, and Databricks**. Programming background in **Python, SQL, Scala, Java, and R**. Experienced managing projects using **Agile, Scrum, and Waterfall methodologies**, and proficient in version **CI/CD** tools such as **Git, Jenkins, and Jira**

Technical Skills

- **Programming Languages:** Python, SQL, R, Java, NoSQL, Shell, .NET
- **Web Technologies:** REST API, Node.JS, ReactJS, HTML, CSS, jQuery, Bootstrap
- **Frameworks & Tools:** NumPy, Pandas, PyTorch, TensorFlow, Sklearn, Seaborn, Matplotlib, Flask, Django, Git, Docker, CI/CD, Statistics, Agile, SAS, Hadoop, Hive, PySpark, Snowflake, Apache (Airflow, Kafka), Linear Regression, Decision Trees, Random Forest, Data Dog, Terraform, Jenkins, Jira, Confluence, Visual Studio, PyCharm
- **Databases:** MS-SQL, MySQL, Redshift, Snowflake, Big Query, PostgreSQL, MongoDB, Cassandra, Informix, IBM DB2, SparkSQL, Oracle, Azure, AWS
- **Data Analysis / ETL:** Power BI, Tableau, Sterling Integrator, Data Bricks, MS Excel, Google Analytics, Machine Learning, Data, Modeling, Data Mapping, Data Mining, Data Extraction, Transformation, Informatica

Professional Experience

Data Engineer, Philips

06/2024 – Present | Remote, USA

- Orchestrated an ETL pipeline using **Azure Data Factory, Apache Airflow, Python, and Apache Kafka**, cutting data processing time by 50%, and improving accuracy, scalability, and efficiency.
- Migrated required enterprise **on-premises databases** to **Azure cloud** using **data migration** agents on top of **Azure Data factory** ensuring minimal downtime and connected to source and sink containers using **Linked lists** to the designed containers in **Azure Data Lake Storage Gen2 (ADLS Gen2)**.
- Utilized **Databricks** notebooks for **advanced data** analysis, feature engineering, and actionable insights from **complex datasets**. Leveraged **Databricks, PySpark, SQL, and Python** distributed computing to process large data volumes effectively.
- Utilized **Azure Databricks** for real-time data processing and analytics enabling **Agile decision-making** and strategic planning to enhance operational efficiency, handling data streams of **1 TB per week**.
- Designed and implemented efficient **data archiving** and **retention** strategies utilizing **Azure Blob Storage** and leveraging **Snowflake's Time Travel** feature, ensuring **optimal data management** and **regulatory compliance**.
- Led the extraction and migration of certain critical data onto **Snowflake** on top of **Azure** depending on the requirement of the business users and clients, ensuring seamless data transfer and storage for enhanced analytics and reporting capabilities.
- Deployed a scalable **real-time data-streaming** platform with **Apache Kafka** and **Apache Spark**, processing over 500GB to 1TB of data daily, boosting data ingestion and easing latency, enhancing **real-time analytics** ability.
- Worked with various data formats like **JSON, XML, CSV, and ORC** and converted them to **Parquet** or **Avro** depending upon the requirements. Worked with **REST** and **SOAP APIs** for data loading, and scaling of pipelines.
- Collaborated with data scientists to integrate **machine learning** models using **TensorFlow** and **Scikit-Learn** into data pipelines, streamlining model deployment, reducing training time by 15% to 20%, and enhancing **predictive analytics** capabilities for improved **business intelligence**.
- Implemented robust data quality checks and monitoring mechanisms using **Apache NiFi** and **Azure Monitor**, ensuring the integrity and availability of data, processing **500 to 1000 MB** records per hour, and maintaining industry standards.
- Ensured data security and privacy by implementing encryption, access controls, and identity management using **Azure Active Directory** and **Azure Key Vault**.
- Utilized **Git** and **GitHub** for version control and source code repositories, enabling efficient collaboration and tracking of code changes in team projects.

Data Engineer, Kroger

07/2023 – 05/2024 | Cincinnati, OH

- Engineered and maintained high-throughput, scalable data pipelines using **Big Query, Kafka, and Azure Event Hubs**, enabling **near-real-time** data processing and efficiently handling up to 42,000 messages per second.

- Led **end-to-end ETL** processes and conducted robust data validation with **PySpark, Python, and Spark SQL**, reducing **data processing** time by 30% and achieving 90% **data accuracy** for enhanced **data quality** and **performance**.
- Administered **Delta Lake tables** within **Azure Data Lake Storage (ADLS) Gen2**, **optimizing data storage** for over 10 TB of **transaction data**, **improving data availability** to 90%, and cutting storage costs by 20%.
- Deployed real-time data streaming solutions with **Delta Lake, Python, and Python**, utilizing **readStream** and **writeStream** for each **Batch** for seamless data processing and high usability at high message rates.
- Integrated vast **logging** and **monitoring** solutions with **Datadog** and **ServiceNow**, enabling automated **tracking** of pipeline **success** and **failure**, easing **incident** response, and improved operational efficiency.
- Managed code **deployments** across **non-production, staging, and production** environments using **Databricks Asset Bundles** and **GitHub workflows**, for consistency, reliability, and version control for robust application delivery.
- Collaborated with **84.51's data scientists** and **BI teams** to develop **Power BI dashboards** that leverage data from the **Unity Catalog**, enhancing the **organization's** ability to make **informed decisions**.

Data Engineer, T-MOBILE06/2022 – 05/2023 | Frisco, Texas

- Directed the migration of 2 TB of customer payment data from **Mainframe** and **Cassandra** to **Azure Data Lake Storage (ADLS)** using **Azure Data Factory**, achieving a 99.9% success rate, enhanced **data accessibility** and **analytics** ability.
- Processed data using **Databricks** by removing 100,000 empty records, flattening hierarchical structures, and replacing null values, resulting 50% gain in **data retrieval times** via efficient **data structuring**.
- Boosted ad **hoc query** performance by 30% via **advanced aggregations** and **complex queries** with **Synapse Analytics**, applying **performance-tuning strategies** to **accelerate data analysis** and **reporting**.
- Developed **complex data transformations** using **Azure Databricks notebooks** and **Synapse analytics** enabling accurate data insights for business stakeholders.
- Enhanced **data security** by **implementing** encryption with **Key Vault**, access controls with **IAM**, and **compliance** measures using **Security Center**, leading 20% drop in security incidents.
- Automated **data workflows** using **Logic Apps, Functions, and Apache Airflow** to streamline **data pipelines** and system interactions, for data movement, process validation, and efficiency of **data tasks**.
- Created 10+ interactive **dashboards** and 50+ reports with **Power BI**, delivering actionable insights and improving decision-making, facilitating **data-driven strategies**, and enhancing user accessibility.

Data Engineer, Collins Aerospace01/2020 – 07/2021 | Hyderabad, India

- Established and maintained an **ETL framework** with **Informatica**, processing over 10 TB of data monthly and streamlined data transformation and processing, reducing processing time by 30%.
- Created and optimized over 50 **ETL workflows** using **Informatica**, integrating data from 15+ diverse sources with 99.9% accuracy, and leveraged **Azure Data Factory** to **orchestrate** and enhance pipeline efficiency.
- Designed and implemented 20+ data models and quality checks within **Informatica**, improving data quality and governance, and utilized **Informatica** features to enhance data management and achieve a 98% data accuracy rate.
- Developed **Informatica workflows** for **real-time data analytics**, reducing reporting latency by 50% and implemented advanced analytics techniques to provide actionable insights within minutes.
- Directed the migration of 5+ on-premises **databases** to **Azure SQL Database**, improving **scalability** and **flexibility**, and used **Informatica migration** tools in conjunction with **Azure SQL Database** for smooth **transition** and **performance**.

Data Analyst, Shree Enterprises06/2018 – 12/2019 | Hyderabad, India

- Directed source-to-target mapping for **ETL tasks** and executed **data profiling** and **documentation**, reducing reporting errors by 45% and ensuring **accurate data conversion**.
- Utilized **Python libraries** (e.g., **Pandas, NumPy**) for data manipulation and analysis, developing scripts to automate data **cleaning** and **visualization**, enhancing **data processing efficiency** and **accuracy**.
- Applied SQL for complex data queries and transformations, using methodologies like **SQL joins, subqueries, and CTEs** for effective data extraction and reporting, improving data insights.
- Created **data ingestion** modules using **AWS** for Loading data in various layers in **S3** and **Redshift**.
- Designed and implemented advanced visualizations in **Power BI**, creating interactive **dashboards** and reports to deliver actionable insights and **support data-driven decision-making**.
- Lead **analysis** suite development writing optimized **SQL queries**, and **triggers** and **build Tableau Dashboards**.
- Create parameters and SSM documents using **AWS Systems Manager**.

Education

Master of Science, Oklahoma State UniversityAug 2021 – May 2023 | Oklahoma, USAComputer Science

Bachelor of Science, Gitam UniversityJul 2016 – June 2020 | Hyderabad, IndiaComputer Science and Engineering

Certification

- AZ-900 Microsoft Azure Fundamentals
- DP-300 Microsoft Azure Data Engineer Associate
- AWS Certified Cloud Practitioner